Machine Learning for Biology and Health

CSCI 1851
Spring 2026

Ritambhara Singh

March 31, 2026
Tuesday

Deep CNNs and New Applications

ChatGPT prompt “minimalist landscape painting a cute penguin working on a computer with DNA on its screen”

Today's goal – Learn about deep CNN architectures and new applications

(1) Deep CNN architectures

(2) Deep CNNs in health applications

(3) Class activity: CNNs for genomics application

(4) New dataset: Single-cell gene expression

Our problem:

Input: $\mathbb{X}$

Pixel Grid

Target: $\mathbb{Y}$

Has Pneumonia?

$x^{(1)} =$

128x128 pixels

Function: f

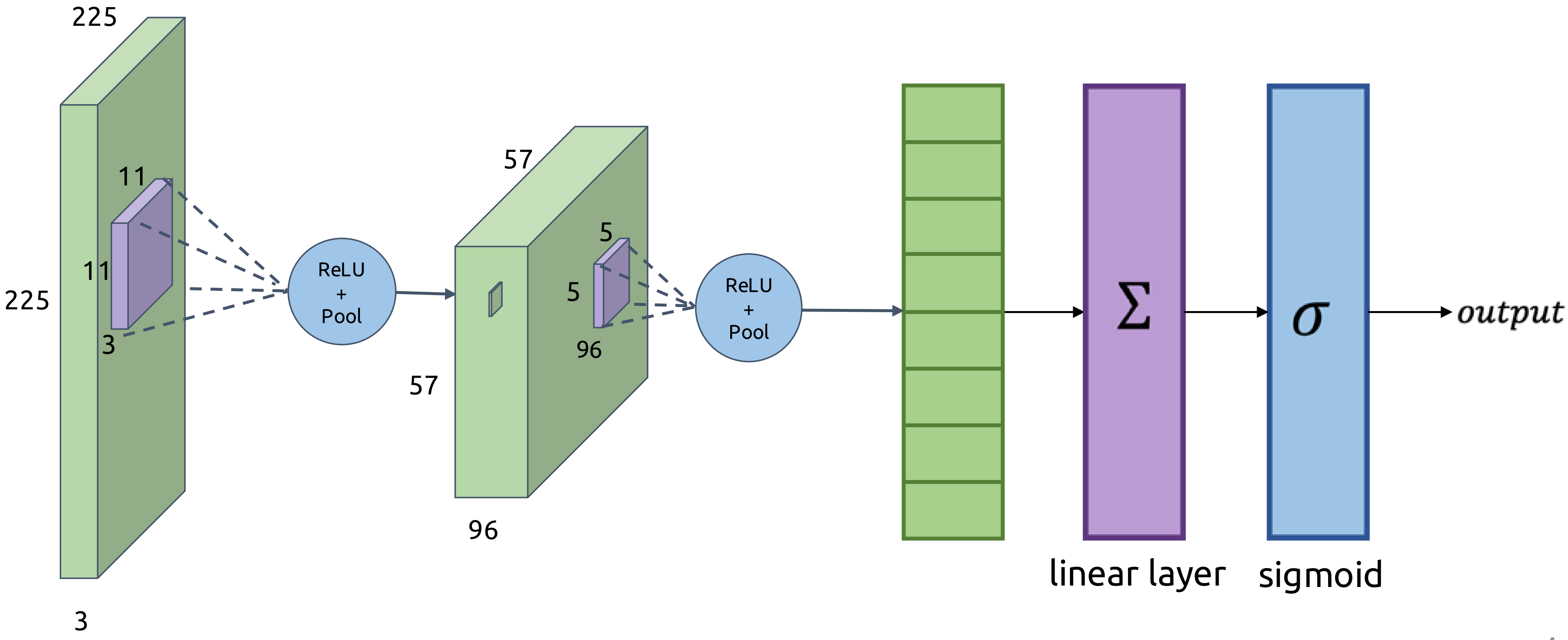
$\boxed{f(\mathbb{X}) \rightarrow \mathbb{Y}}$

$y^{(1)} = 0$

$x^{(2)} =$

$y^{(2)} = 1$

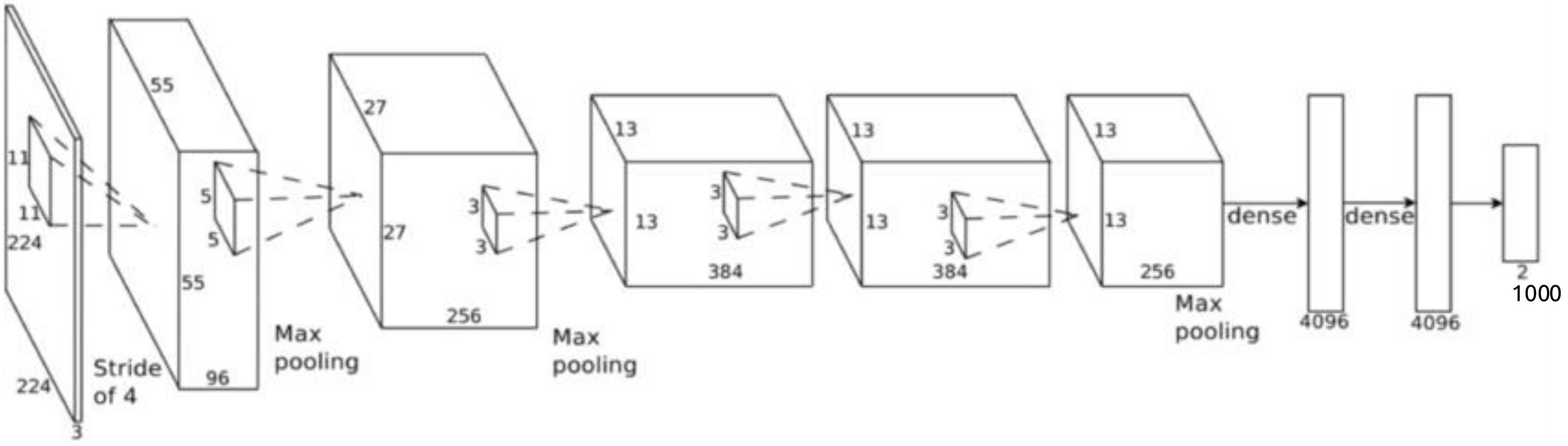
CNN Architecture



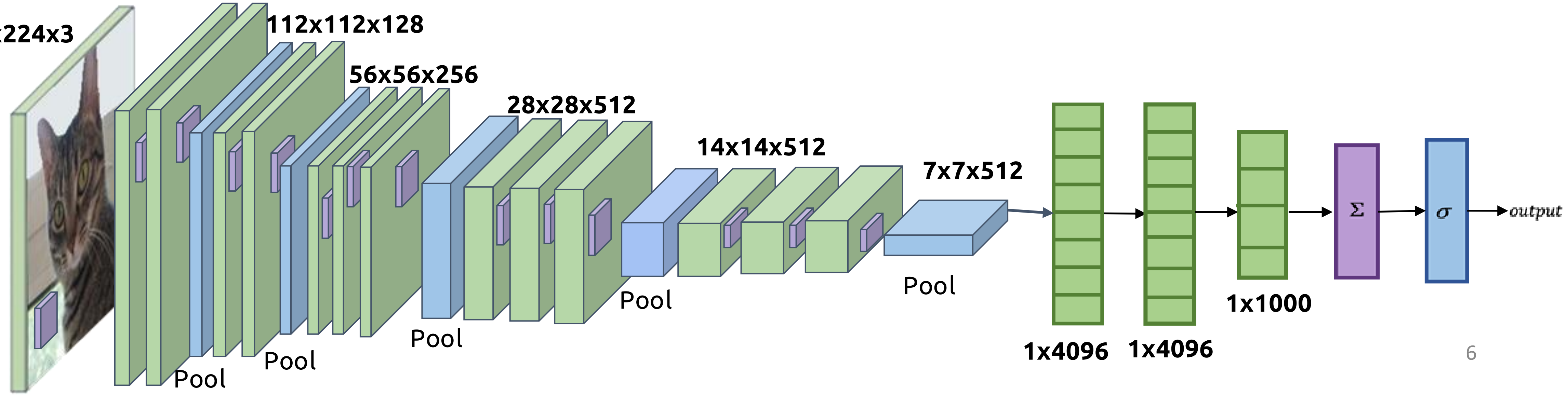
Deep CNN architectures

More Complicated Networks

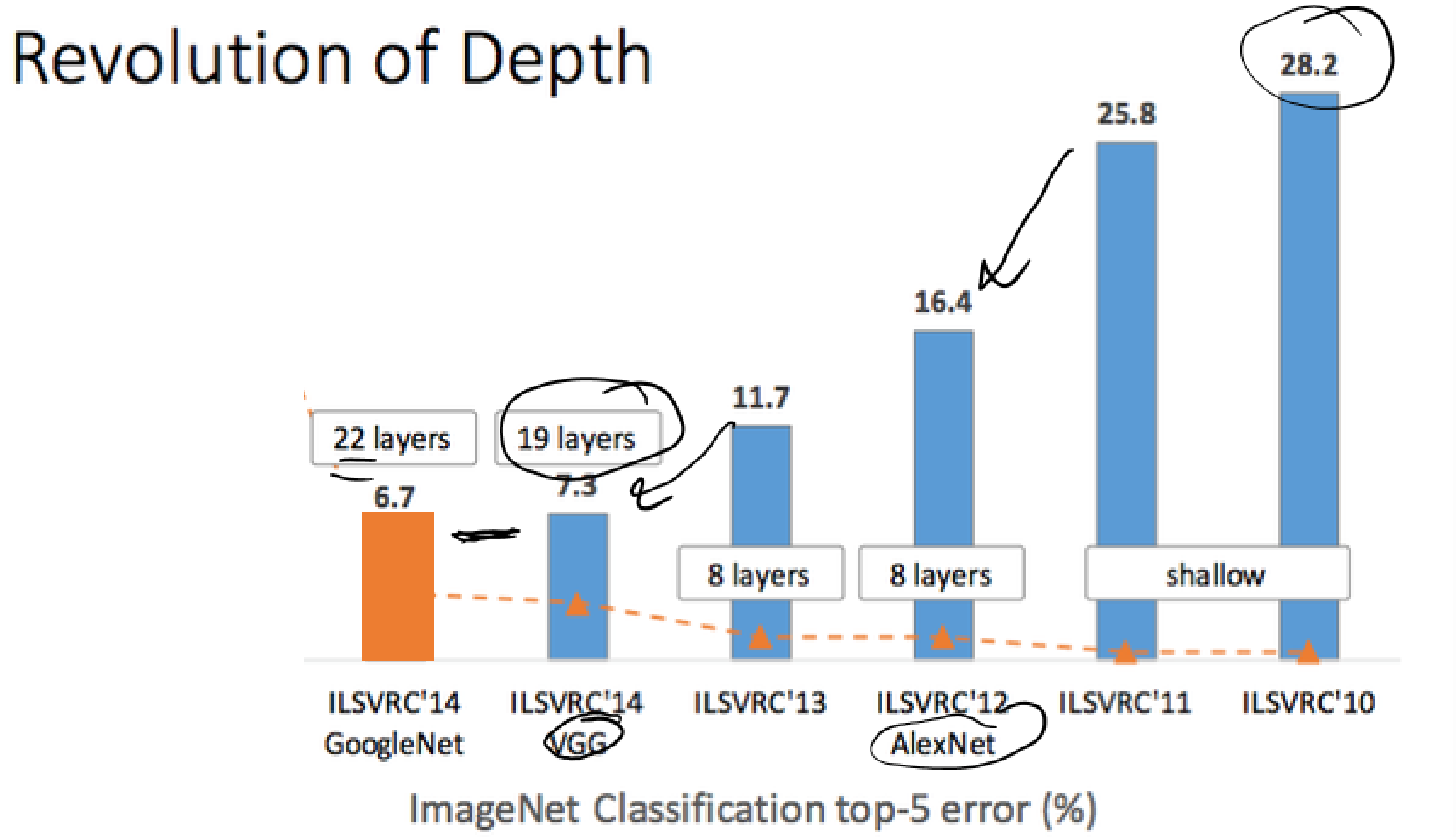
AlexNet:



VGG:



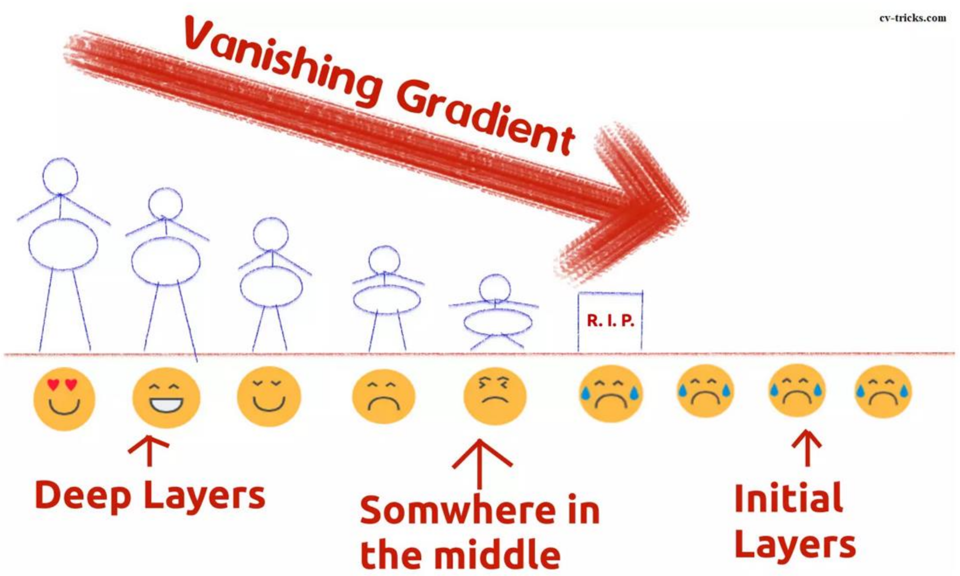
Can you guess what was the biggest bottleneck to adding more layers?

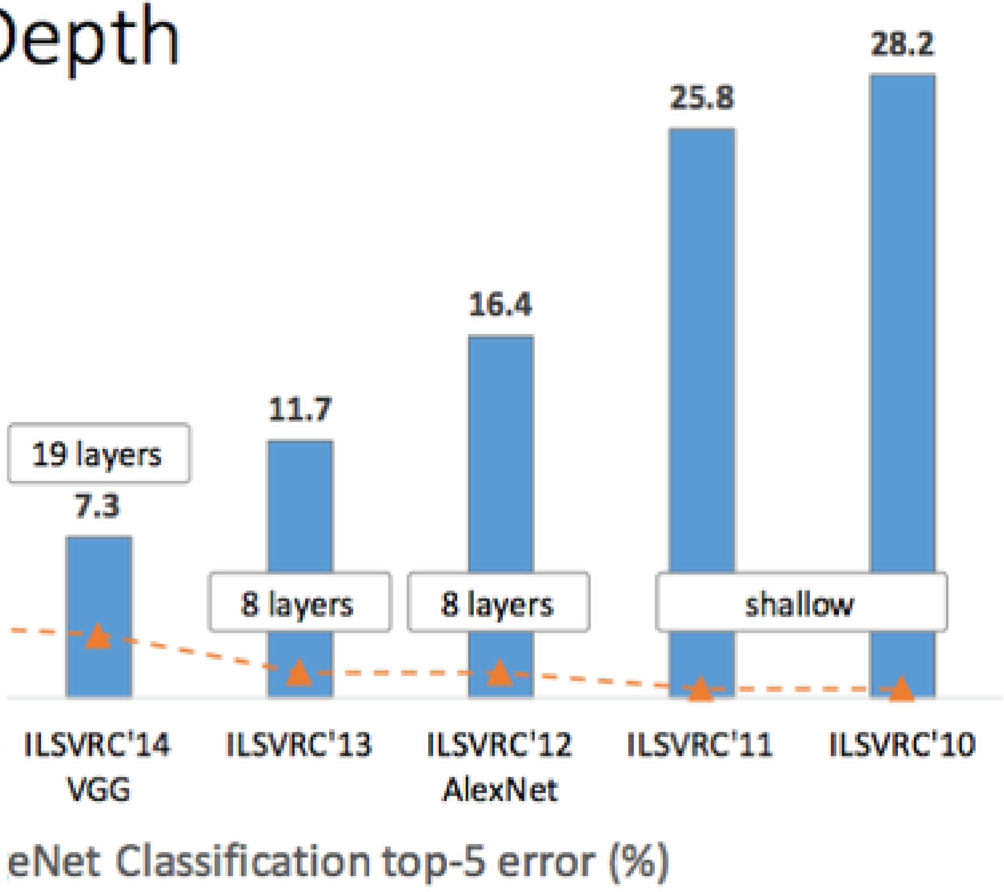


https://towardsdatascience.com/understanding-and-coding-a-resnet-in-keras-446d7ff84d33

Revolution of Depth

Vanishing gradients!



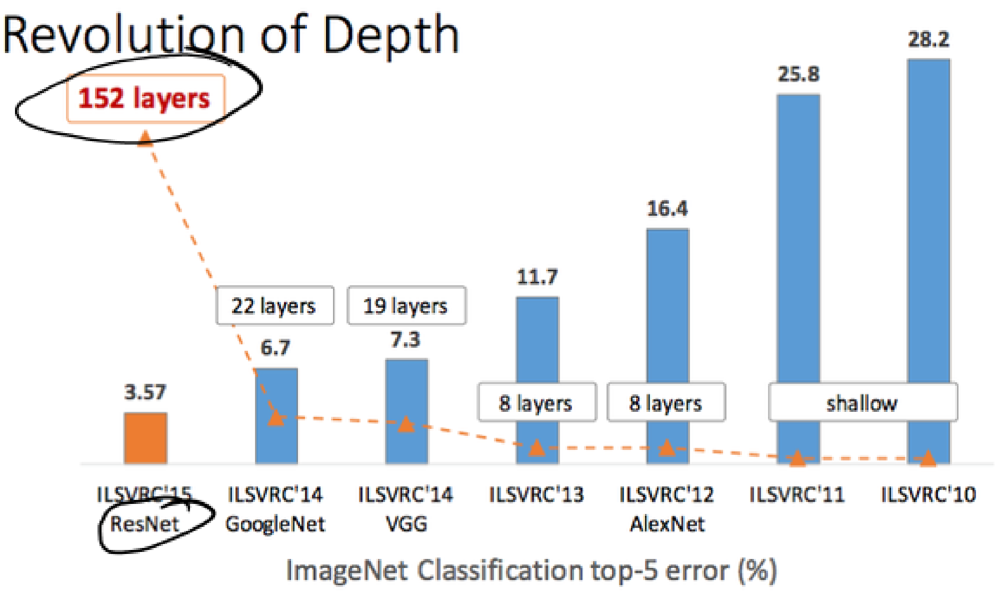


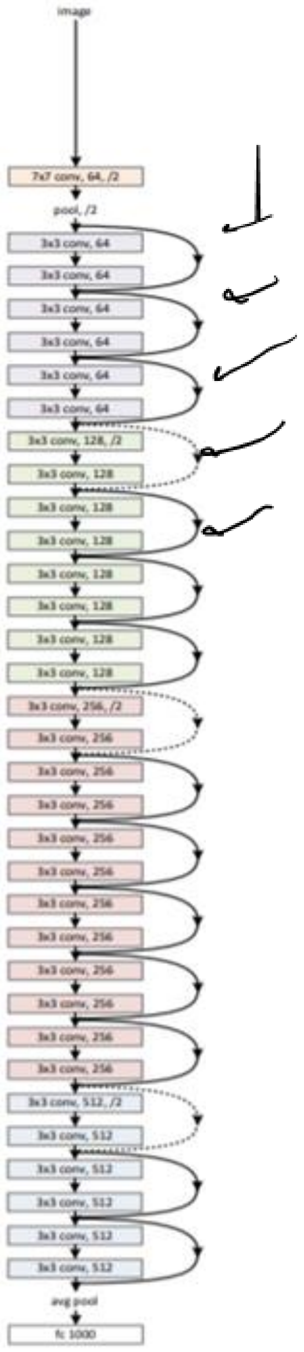
https://towardsdatascience.com/understanding-and-coding-a-resnet-in-keras-446d7ff84d33, https://cv-tricks.com/keras/understand-implement-resnets/

More Complicated Networks

ResNet:

Lots of layers, tons of learnable parameters
Avoids Vanishing Gradient problem
but how?

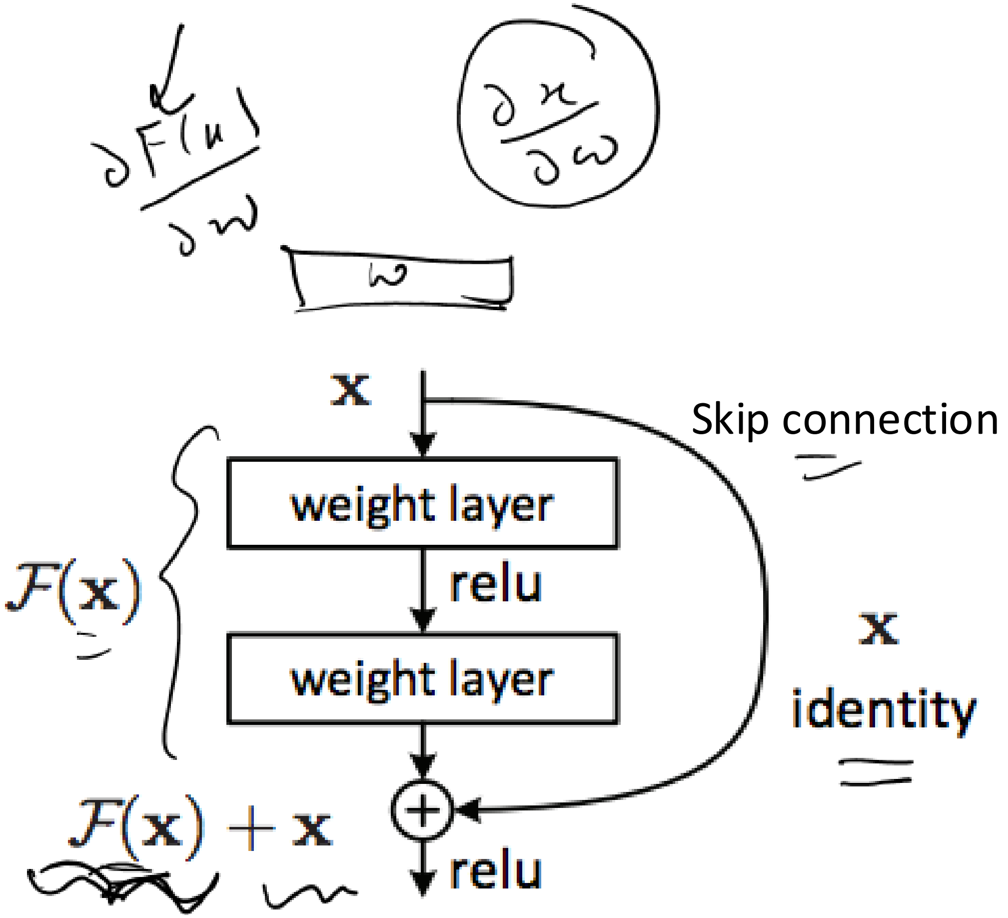




K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. arXiv preprint arXiv:1512.03385, 2015.

Residual Blocks

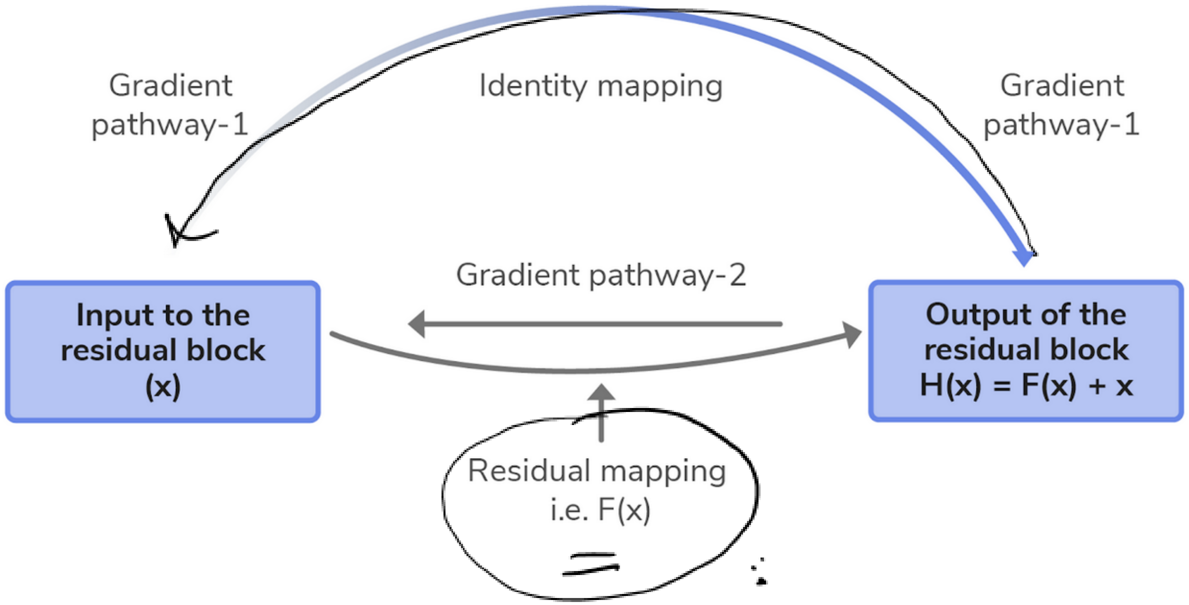
- In very deep nets, each layer often needs to learn just a small transformation of the preceding layer (identity + change)
- Idea: explicitly design the network such that the output of each layer is the identity + some deviation from it
    - Deviation is known as a residual



Any questions?

Residual Blocks

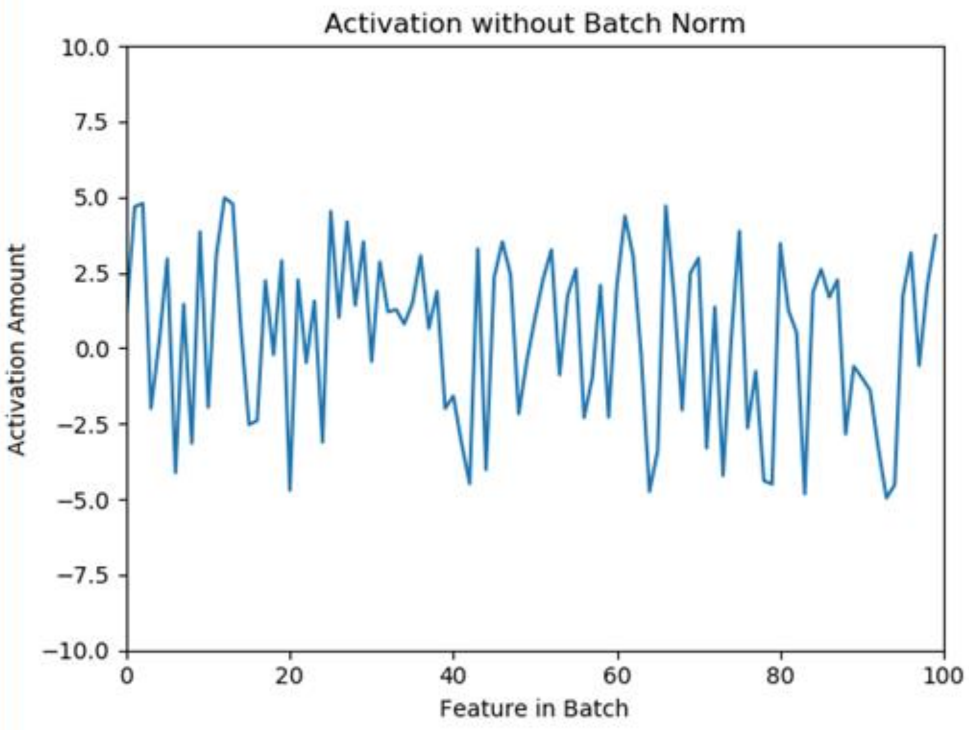
- In very deep nets, each layer often needs to learn just a small transformation of the preceding layer (identity + change)
- Idea: explicitly design the network such that the output of each layer is the ident + some deviation from it
  - Deviation is known as a residual
- Allows gradient to flow through two pathways
- **Significantly stabilizes training of very deep networks**

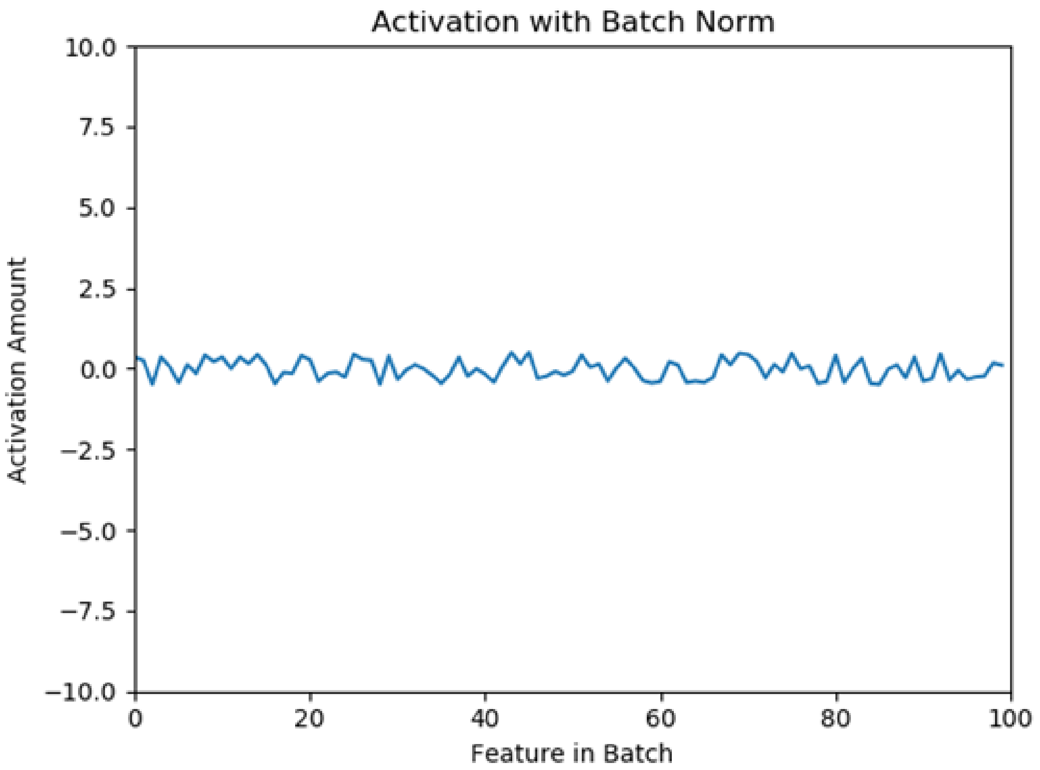


https://blog.perceptilabs.com/using-resnets-to-detect-anomalies-in-industrial-iot-textile-production/

Batch Normalization (stabilizing training)

Idea: normalize the activations for each feature at each layer



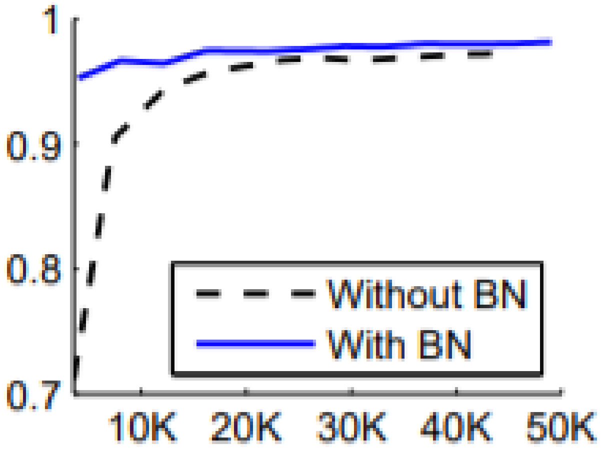


Why might we want to do this?

Batch Normalization: Motivation

More stable inputs = faster training

MNIST test accuracy vs number of training steps



https://arxiv.org/pdf/1502.03167.pdf

Batch Normalization: Implementation

For each feature x, Start by calculating the batch mean and standard deviation for each feature:

$$\mu_{batch} = \frac{\sum_{i=0}^{batch\_size} x_i}{batch\_size}$$

$$\sigma_{batch} = \sqrt{\frac{\sum_{i=0}^{batch\_size}(x_i - \mu_{batch})^2}{batch_{size}}}$$

Batch Normalization: Implementation

Normalize by subtracting feature x's batch mean, then divide by batch standard deviation.

$$x' = \frac{x - \mu_{batch}}{\sigma_{batch}}$$

Feature x now has mean 0 and variance 1 along the batch

Have we seen this before?

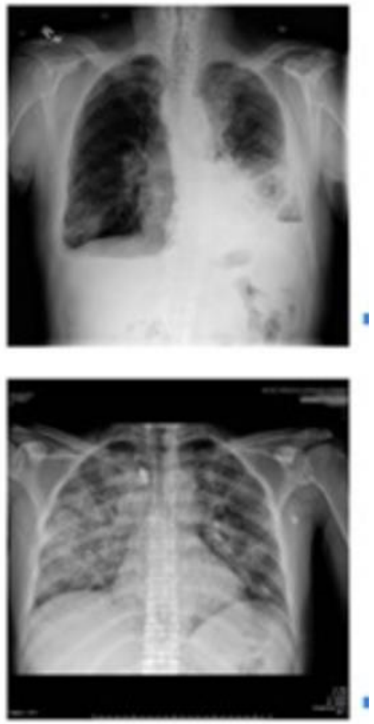
Batch Normalization in Pytorch

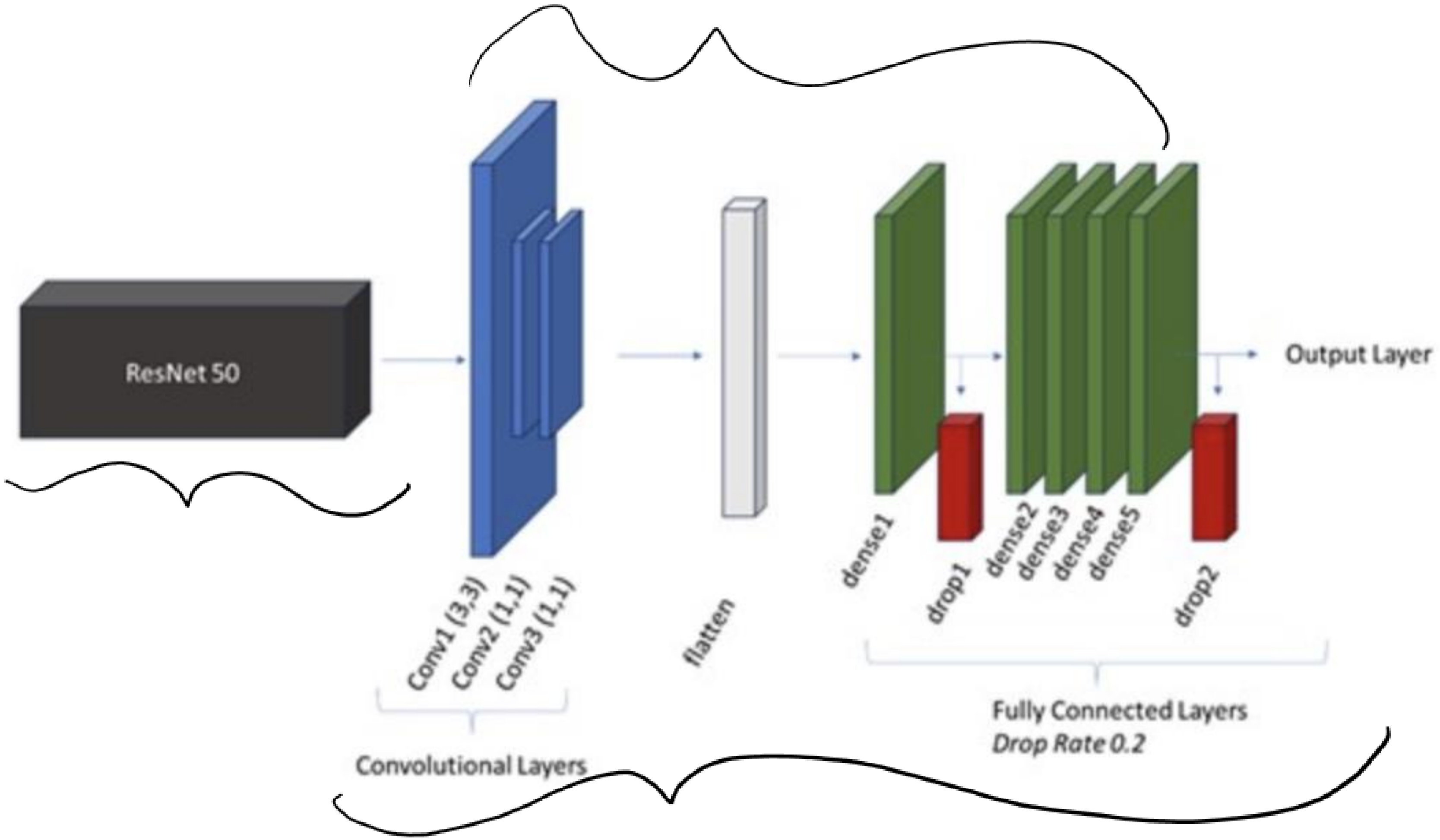
nn.BatchNorm1d(*num_features*)

https://docs.pytorch.org/docs/stable/generated/torch.nn.BatchNorm1d.html

Deep CNNs in health applications

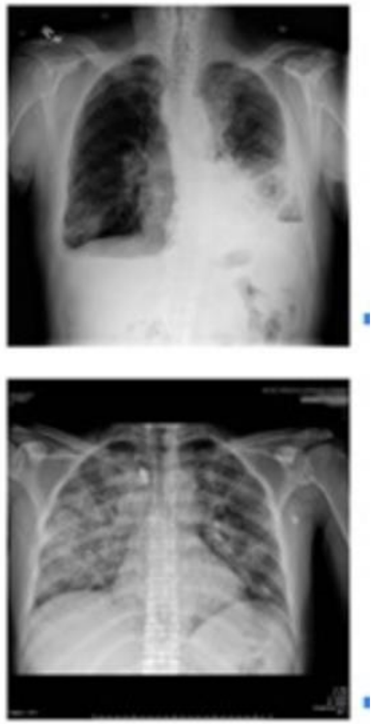
Resnet-based model for classifying chest X-rays as normal or abnormal





https://academic.oup.com/jamiaopen/article/7/2/ooae035/7661115

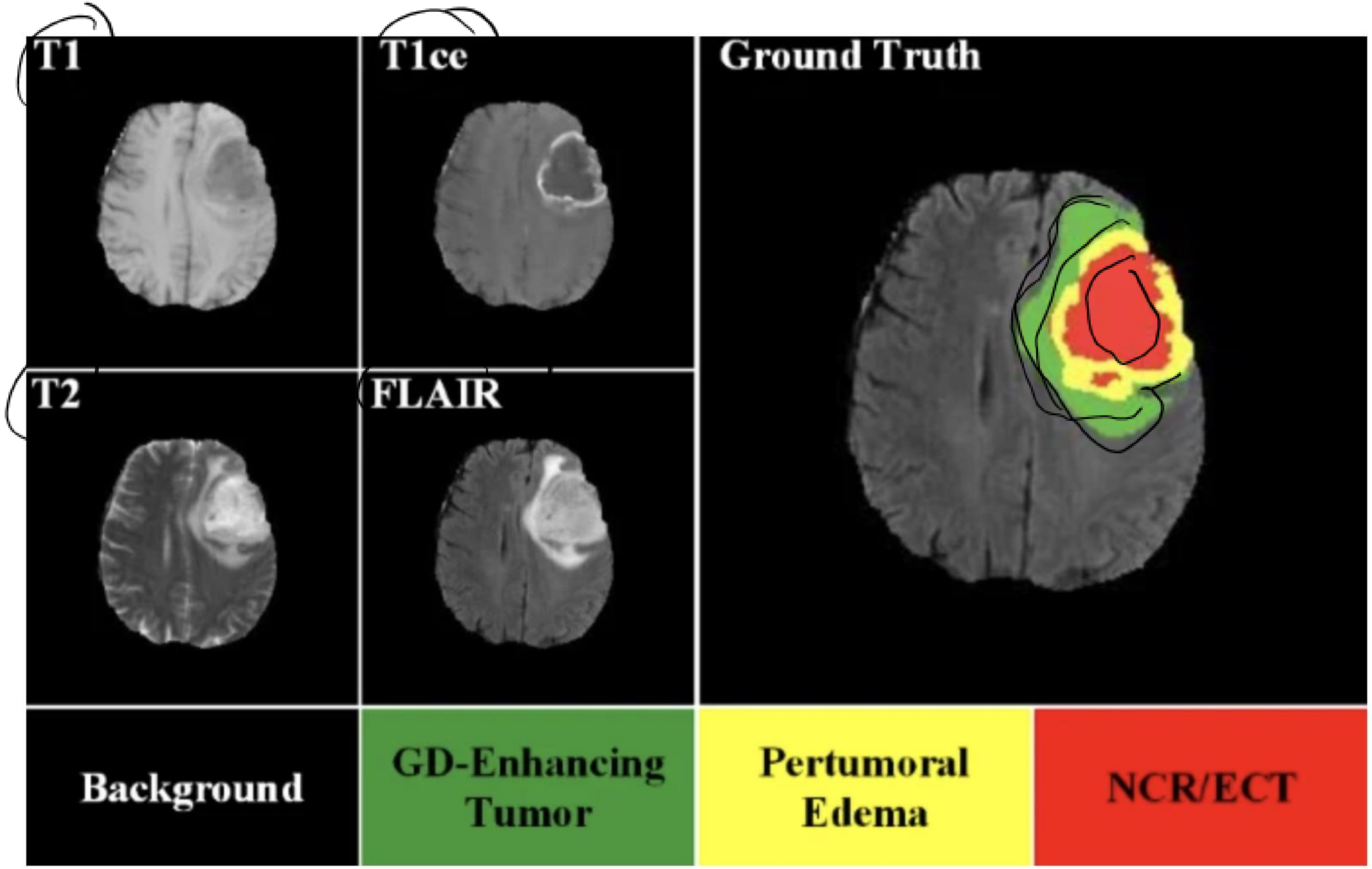
Resnet-based model for classifying chest X-rays as normal or abnormal



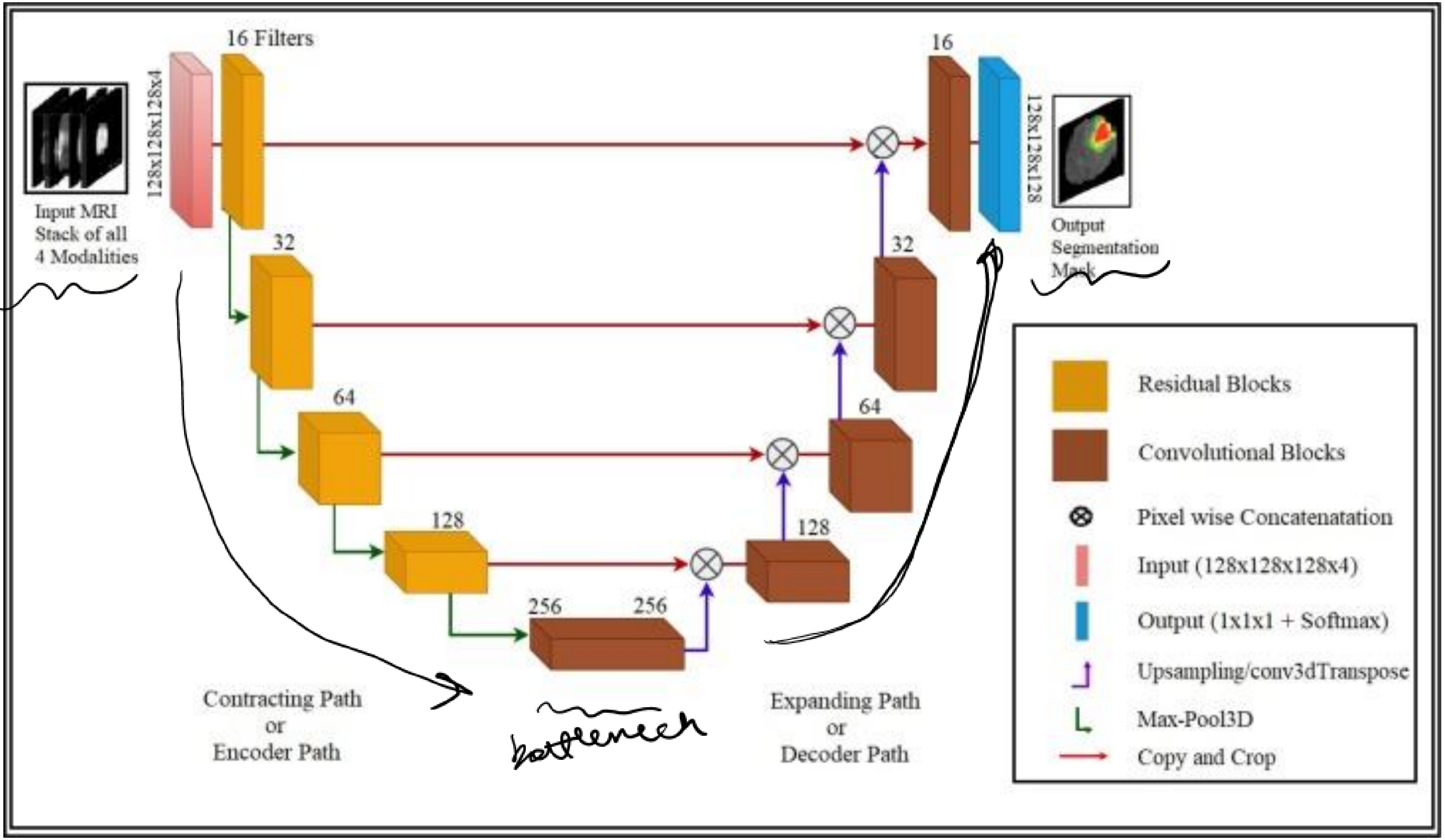
Model	Accuracy	Recall	Precision	AUC
Custom model (ResNet50 Base)	0.7866	0.6873	0.9615	0.9023
With contrast enhancement	0.8190	0.7423	0.9600	0.9020
Custom model (ResNet101 Base)	0.7845	0.7113	0.9283	0.8823
With contrast enhancement	0.7931	0.7354	0.9185	0.8834
Custom model (ResNet152 Base)	0.7672	0.6667	0.9463	0.8664
With contrast enhancement	0.7672	0.6667	0.9463	0.8724
ResNet50 Base (no additional layers or training)	0.3685	0.0103	0.3750	0.2815
With contrast enhancement	0.3685	0.0103	0.3750	0.2812

https://academic.oup.com/jamiaopen/article/7/2/ooae035/7661115

Brain tumor segmentation



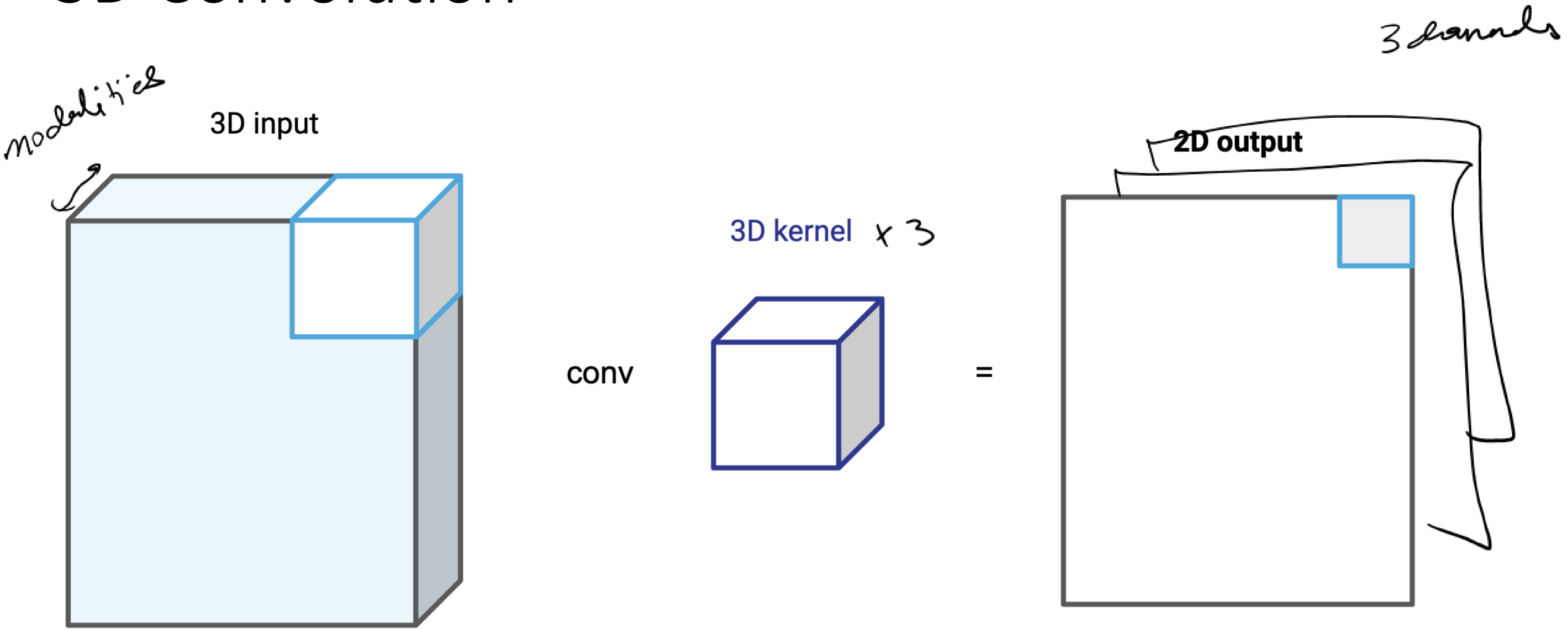
3D U-Net for brain tumor segmentation



https://www.sciencedirect.com/science/article/pii/S1746809422003809

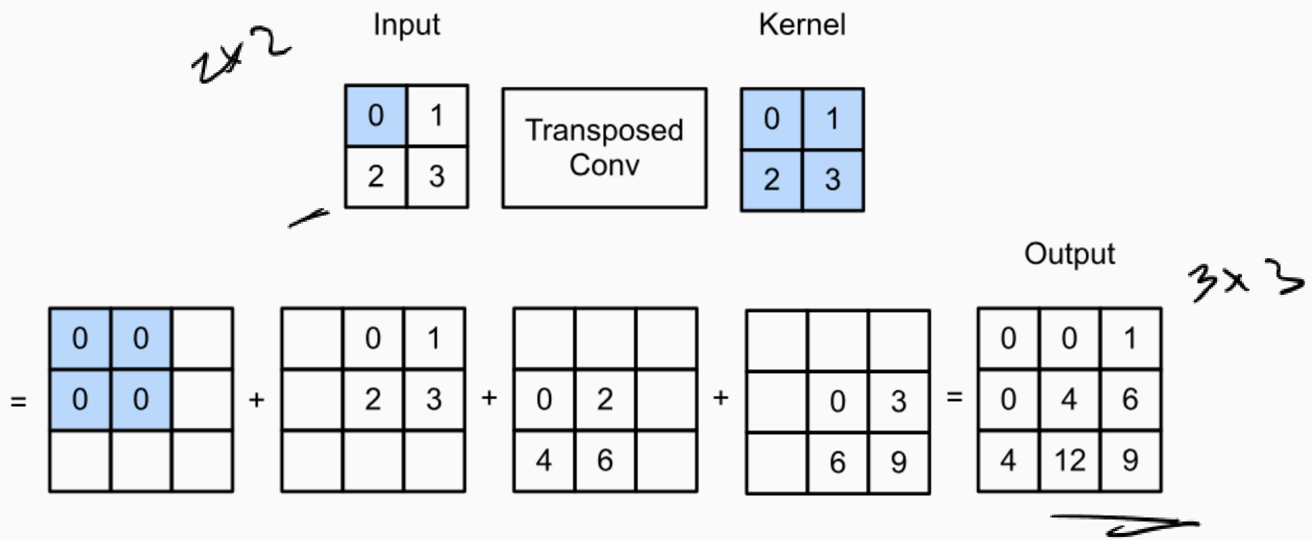
3D Convolution

Why 3D??



Transposed Convolution

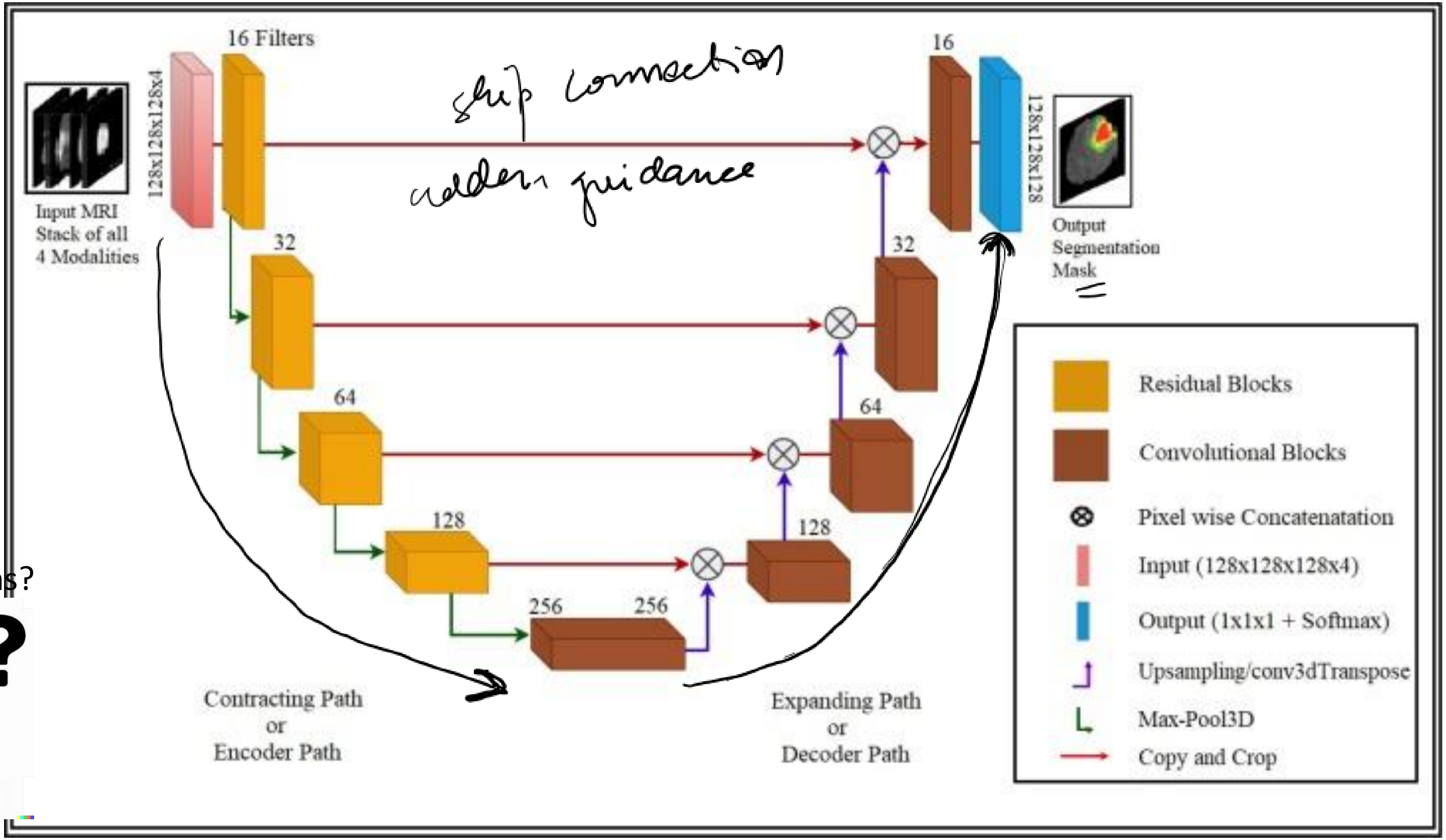
The transposed convolution *broadcasts* input elements via the kernel, thereby producing an output that is larger than the input



Transposed convolution with a 2×2 kernel. The shaded portions are a portion of an intermediate tensor as well as the input and kernel tensor elements used for the computation.

https://d2l.ai/chapter_computer-vision/transposed-conv.html

3D U-Net for brain tumor segmentation



Any questions?

https://www.sciencedirect.com/science/article/pii/S1746809422003809

CNNs for genomics application

How do we view the DNA

What is a gene?

Coordinates: chr1: 14569-14780

GGCGAGTCCGCGGGGAGGCGGCAGAGCCTGAGCTCAGGTCTTTCTGCGTCTGGCGGA GGCGAGTCCGCGGGGAGGCGGG

Gene

FASTA file

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Image courtesy: https://compgenomr.github.io/book/genes-dna-and-central-dogma.html#what-does-a-gene-look-like

Deep Learning for DNA sequence classification

>KM822100.1 |Lassa virus strain LASV1000-NIG-2009 segment L Z protein (Z) and polymerase (L) genes, complete cds
cccttcagtcttaggctacccccacttgact---gaatcataactctttgc--------tttgatttactatagcaaagagcataa---------
cctttgaaatcaatccattctccaagagattcaagaagttcactcagtatgttagcataatccacaggtctctgtcttgtgaggattgagacaatgtccacacctgtccattgaattctttgtt
ttaatctatcagacatcatt……..

>KM822015.1 |Lassa virus strain LASV056-NIG-2008 segment L Z protein (Z) and polymerase (L) genes, complete cds
ccctttagtcttaggctgcccccacttgact---gaatcataactctttgc--------ctcgatttactgtagcaaagagcatag---------
cctttgaaatcaatccattctccaagagattcaagaagctcactcagtatatcagcatgatccacaggtctctgttttgtgaggattgaaacaatgtccacacccgtccattgaatcctttgt
ttcaatctatcggacatcattgatttcaatgcaccaactaaaagatgcggggaactatcaccaatctcccttataaacatt…..

A = 1
T = 2
C = 3
G = 4
N = 5

Labels: 0 (came from location X) and 1 (came from location Y)

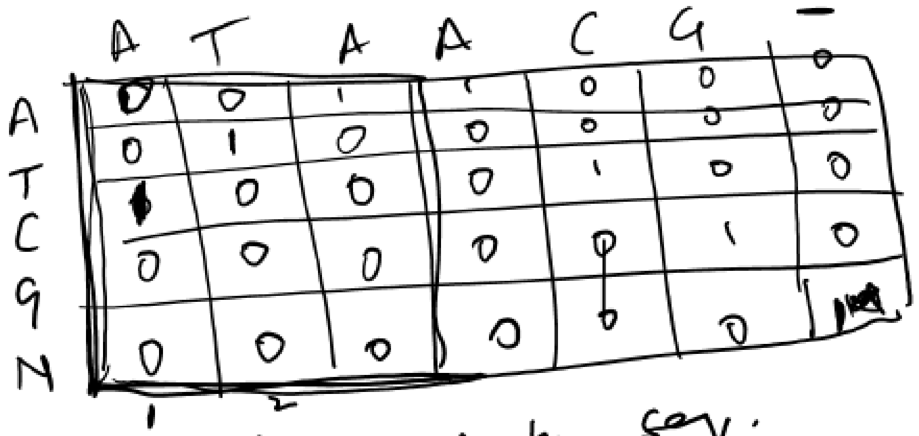
Help your colleague design a deep learning model for this task:
(1) How will you represent the sequences?
(2) What model to use?
(3) Design a basic model with the layers covered so far.
(4) Can we do any interpretation after the model is trained?

Help your colleague design a deep learning model for this task:
(1) How will you represent the sequences?
(2) What model to use?
(3) Design a basic model with the layers covered so far.
(4) Can we do any interpretation after the model is trained?

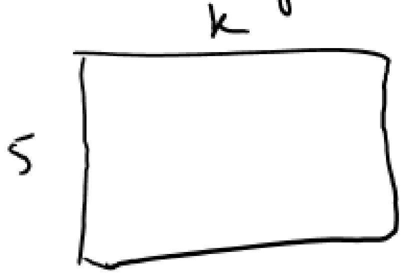
1D convolution

1. Filter weights
2. Activations after 1 conv layer.
3. Perturbation

90%.



length of the seq.



seq → 1D CNN → 1D CNN → Flatten → MLP → Sigmoid

How do we view the DNA

What is a gene?

Can we get any other information about the gene?

Coordinates: chr1: 14569-14780

GGCGAGTCCGCGGGGAGGCGGGCAGAGCCTGAGCTCAGGTCTTTCTGCGTCTGGCGGA GGCGAGTCCGCGGGGAGGCGGG

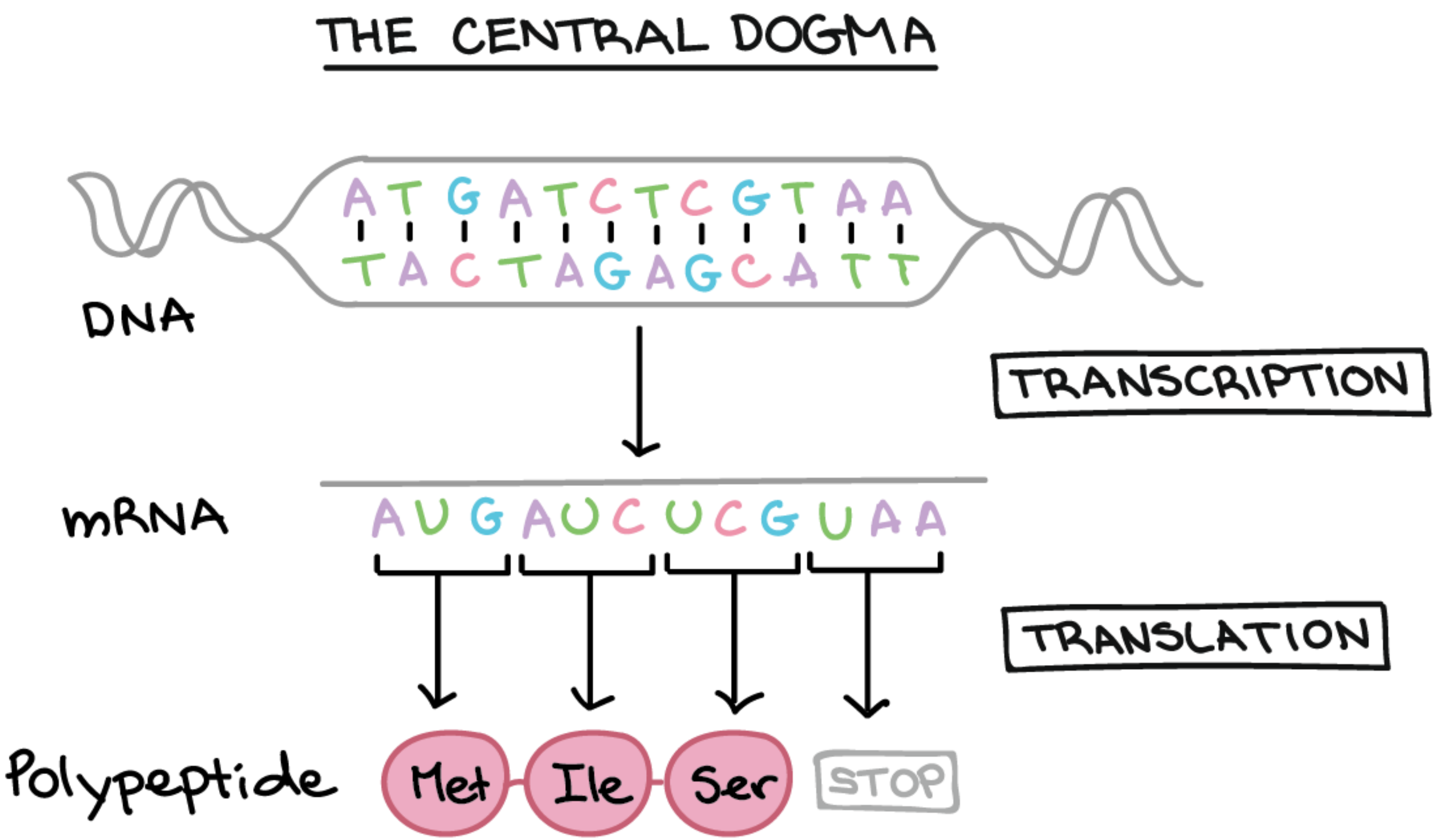
Gene

FASTA file

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Image courtesy: https://compgenomr.github.io/book/genes-dna-and-central-dogma.html#what-does-a-gene-look-like

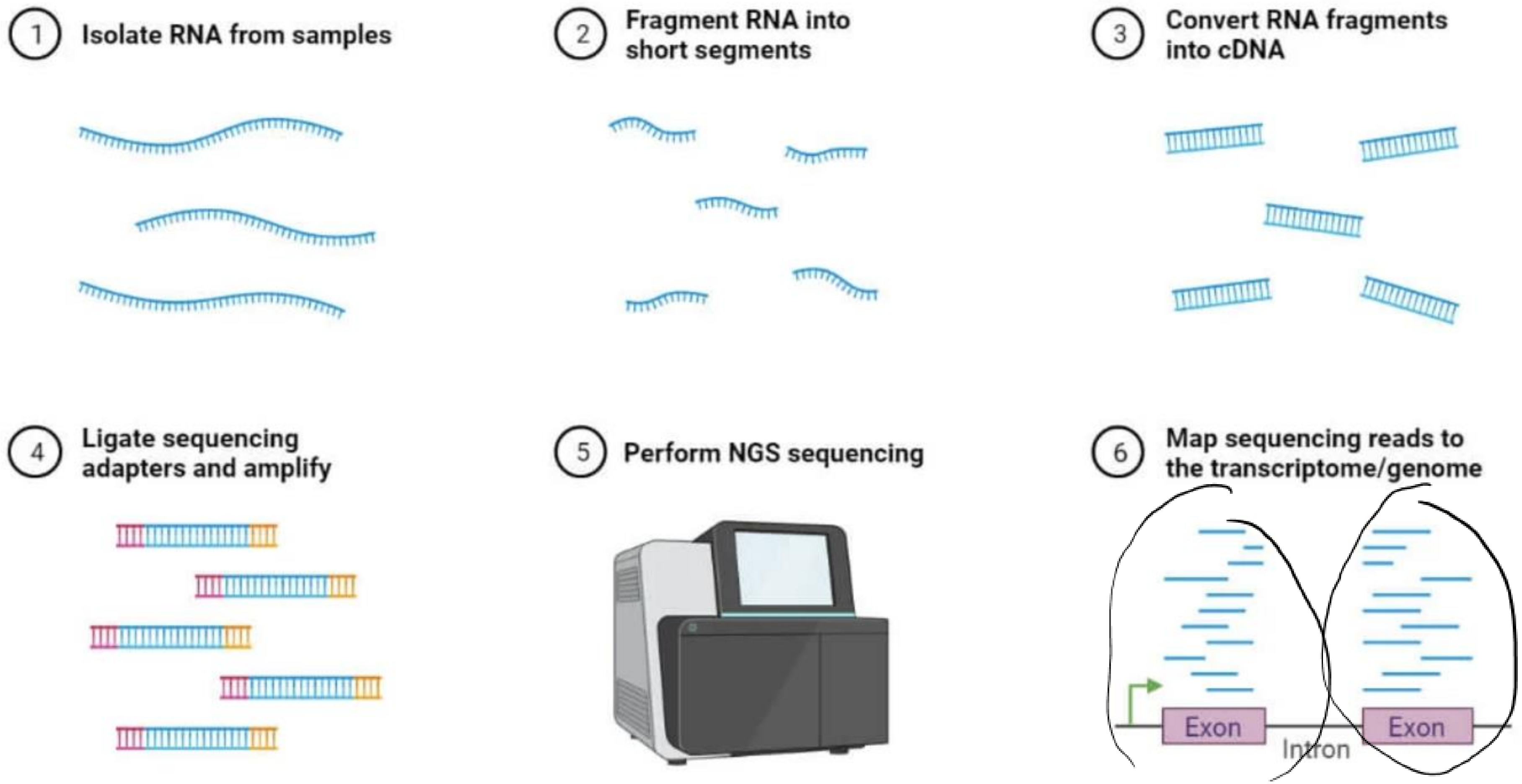
Gene expression



How would you measure gene expression?

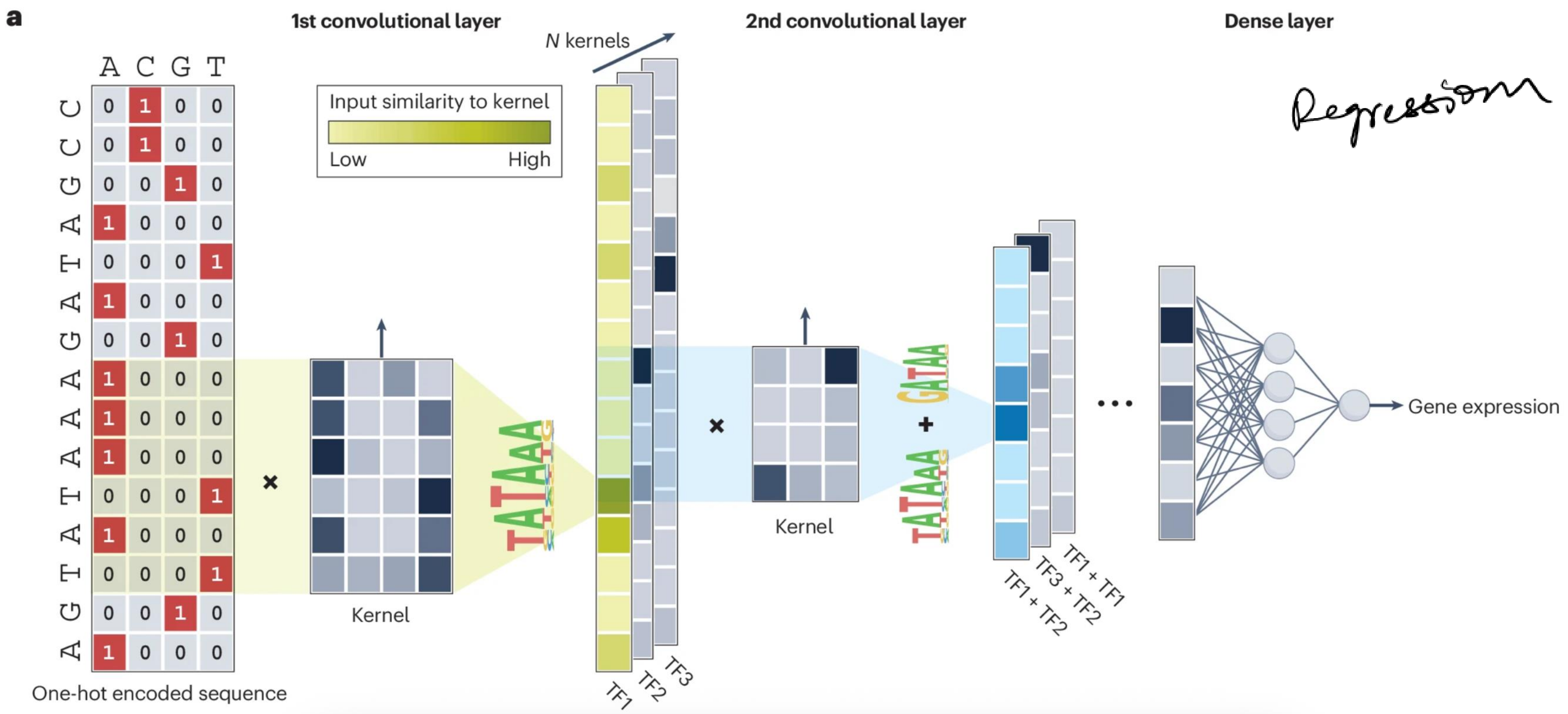
Image courtesy: https://www.khanacademy.org/science/high-school-biology/hs-molecular-genetics/hs-rna-and-protein-synthesis/a/intro-to-gene-expression-central-dogma

RNA Sequencing



https://microbenotes.com/rna-sequencing-principle-steps-types-uses/

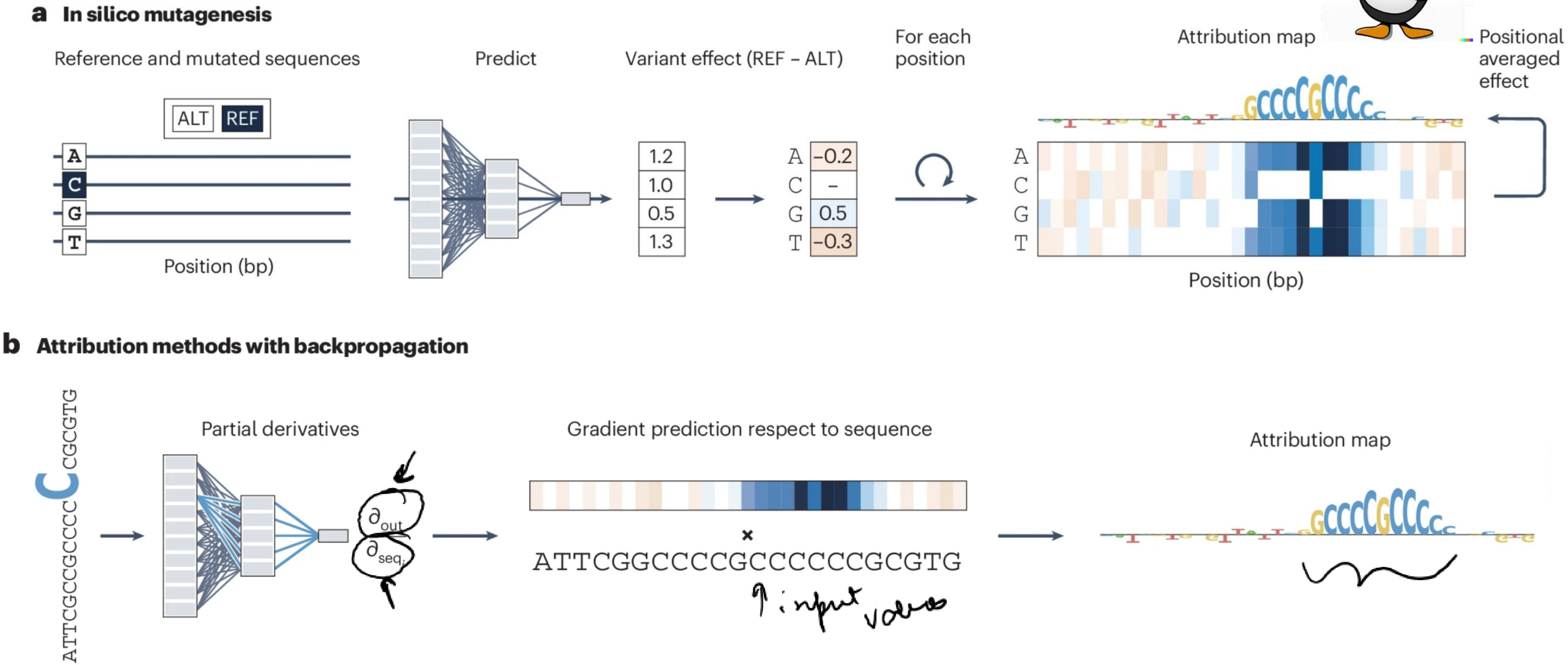
Predicting gene expression from DNA sequence using CNNs



https://www.nature.com/articles/s41576-025-00841-2

Any questions?

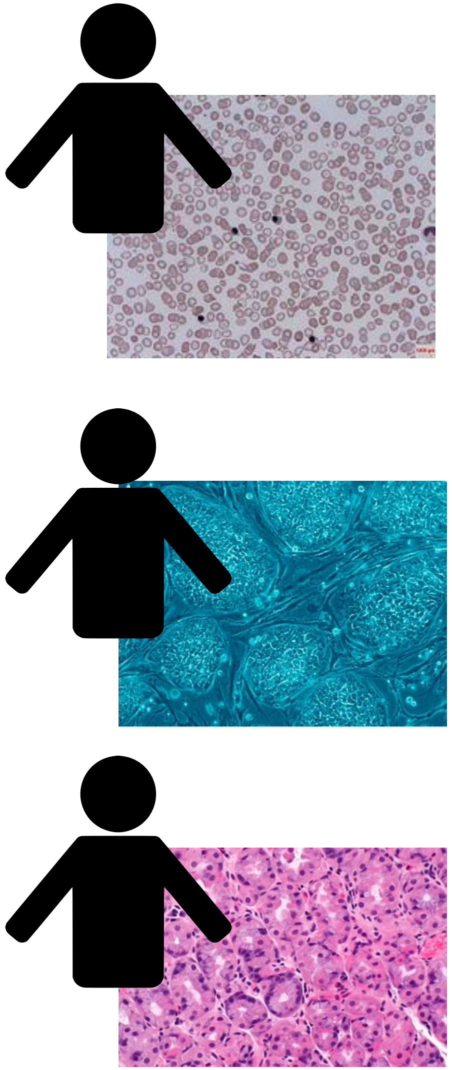
Explaining model predictions



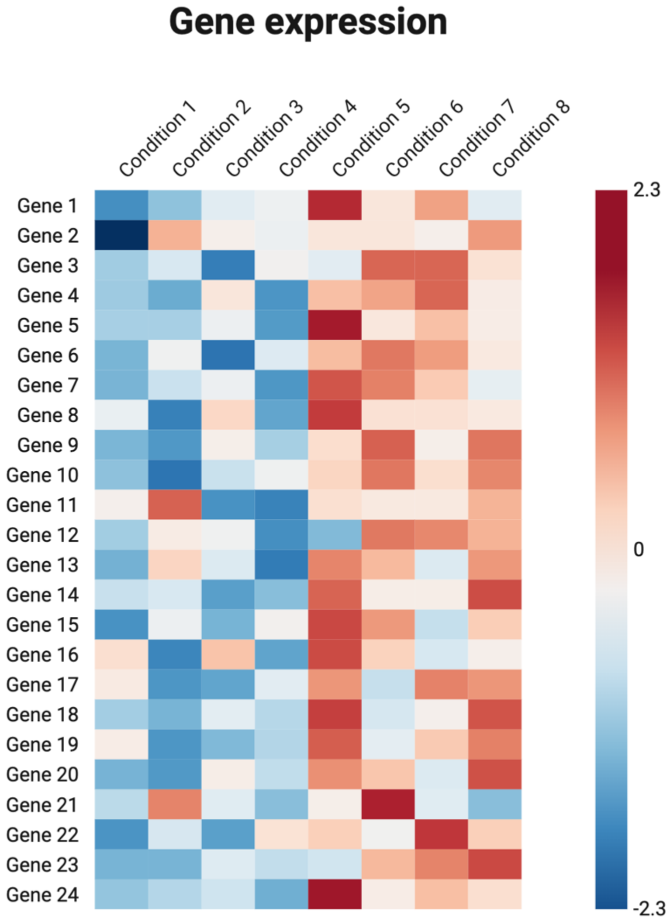
https://www.nature.com/articles/s41576-025-00841-2

New dataset: Single-cell gene expression

Measuring gene expression for different conditions



RNA-sequencing



~20,000 genes!
for humans

Single-cell Data

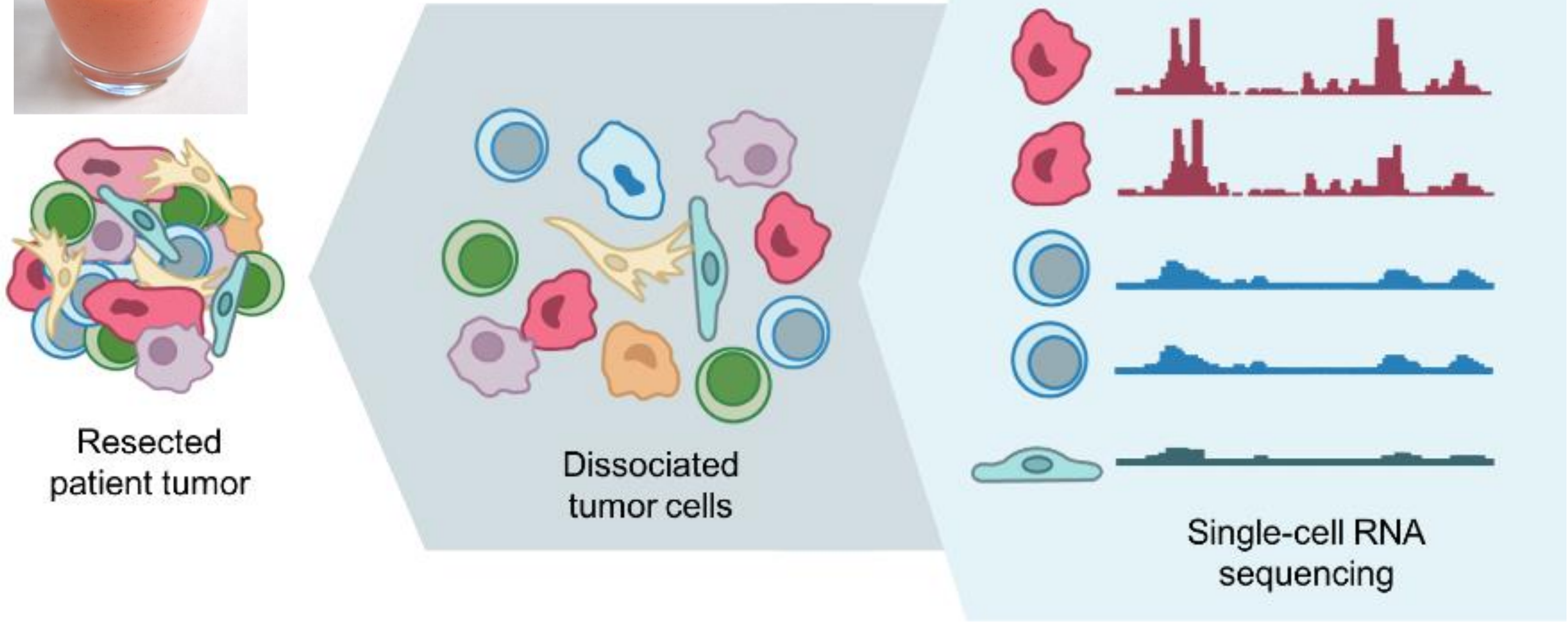
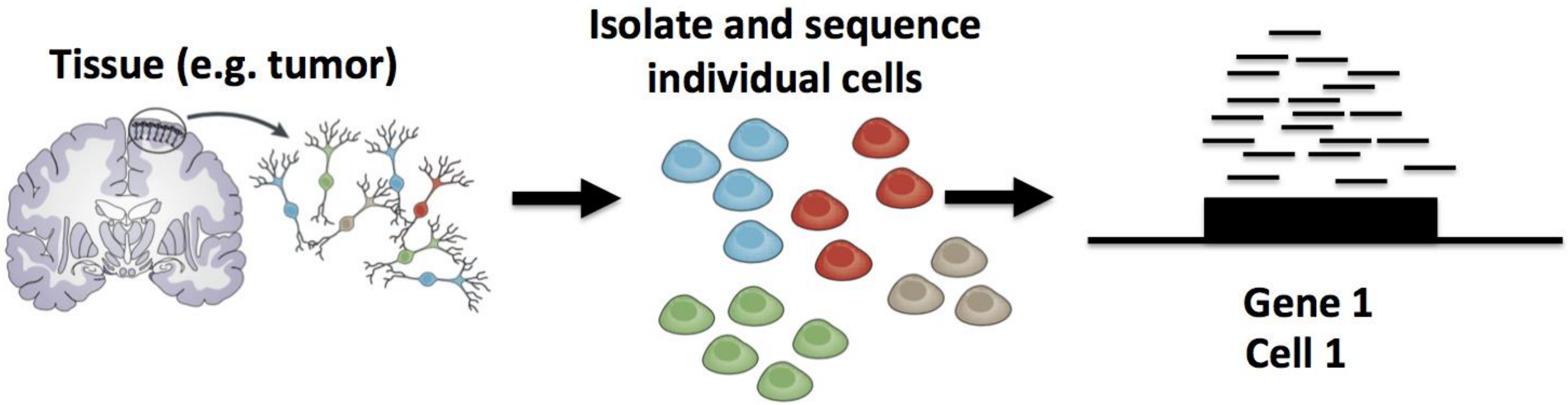


Image courtesy: https://www.cancer.gov/about-nci/organization/ccg/blog/2019/interview-scrna-seq

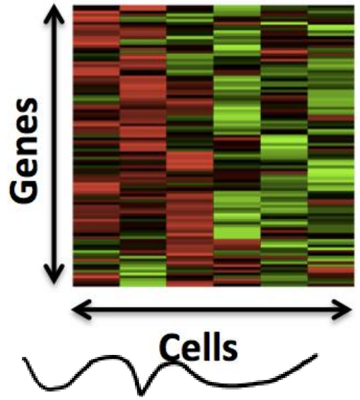
Single-cell RNA-Seq (scRNA-Seq)

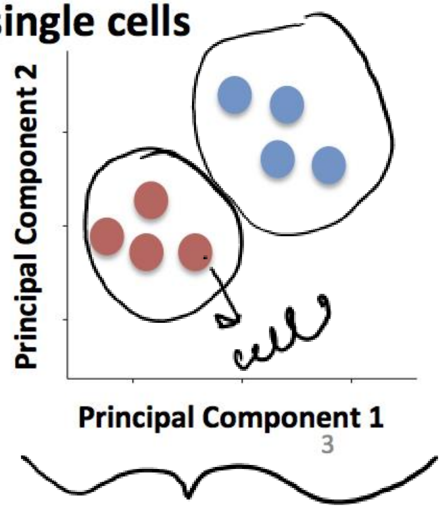


Read Counts

	Cell 1	Cell 2	...
Gene 1	18	0	
Gene 2	1010	506	
Gene 3	0	49	
Gene 4	22	0	
...			

Compare gene expression profiles of single cells





Any questions?

https://learn.gencore.bio.nyu.edu/single-cell-rnaseq/

Challenges of working with single-cell data

- Data

    - Sparsity, errors and missing data
    - "Nuisance factors": Batch effects, etc.

- Downstream analysis:

    - Mapping to reference atlas
    - Integrating measurements

Review | Open Access | Published: 07 February 2020

Eleven grand challenges in single-cell data science

David Lähnemann, Johannes Köster, [...] Alexander Schönhuth

Challenges of working with single-cell data

- Data
  - Sparsity, errors and missing data

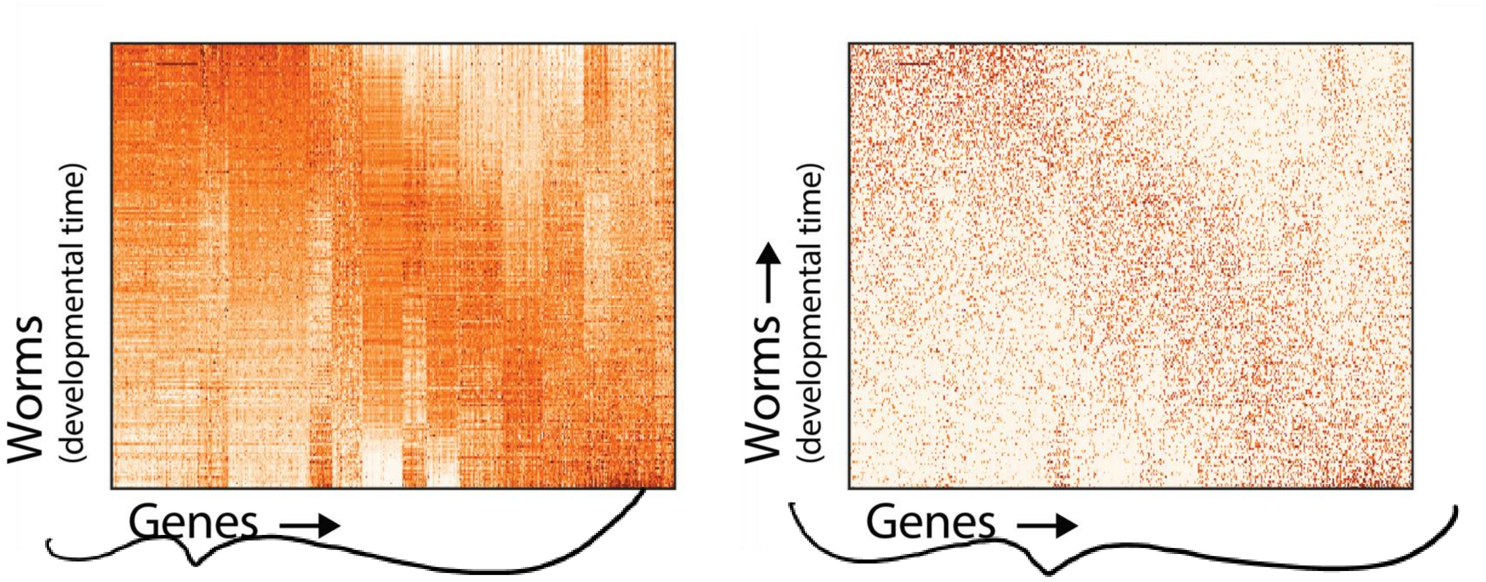
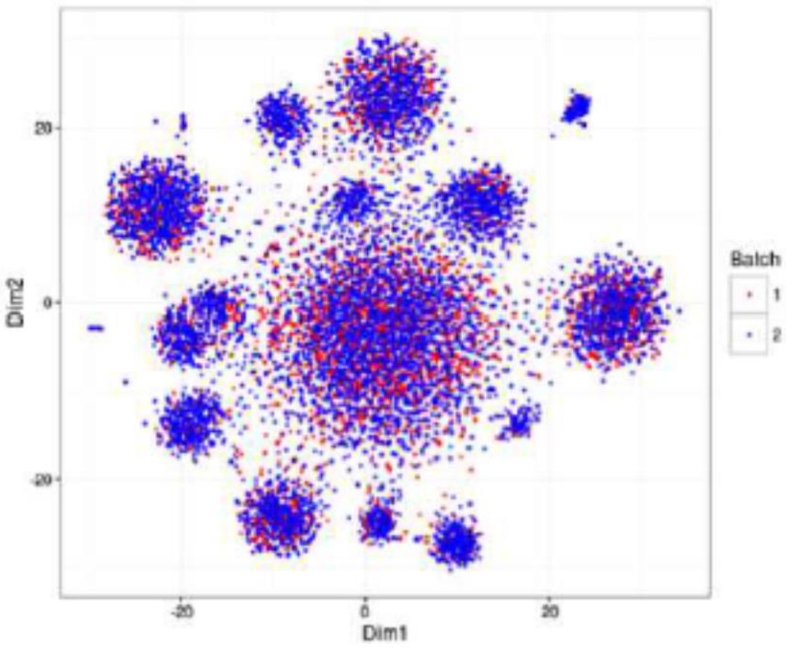


Image courtesy: https://www.krishnaswamylab.org/blog/2018/10/28/denoising-noisy-gene-expression-in-scrna-seq

Challenges of working with single-cell data

- Data
  - “Nuisance factors”: Batch effects, etc.



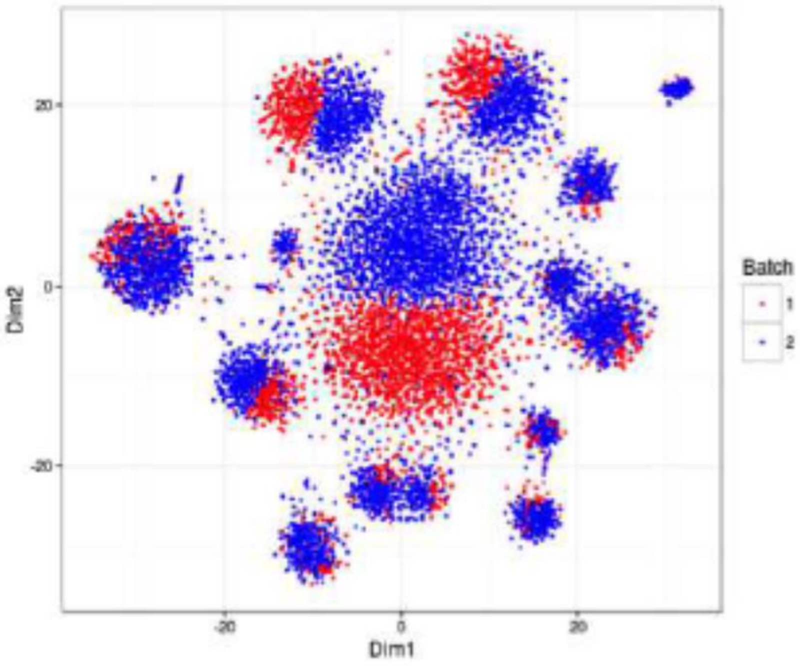
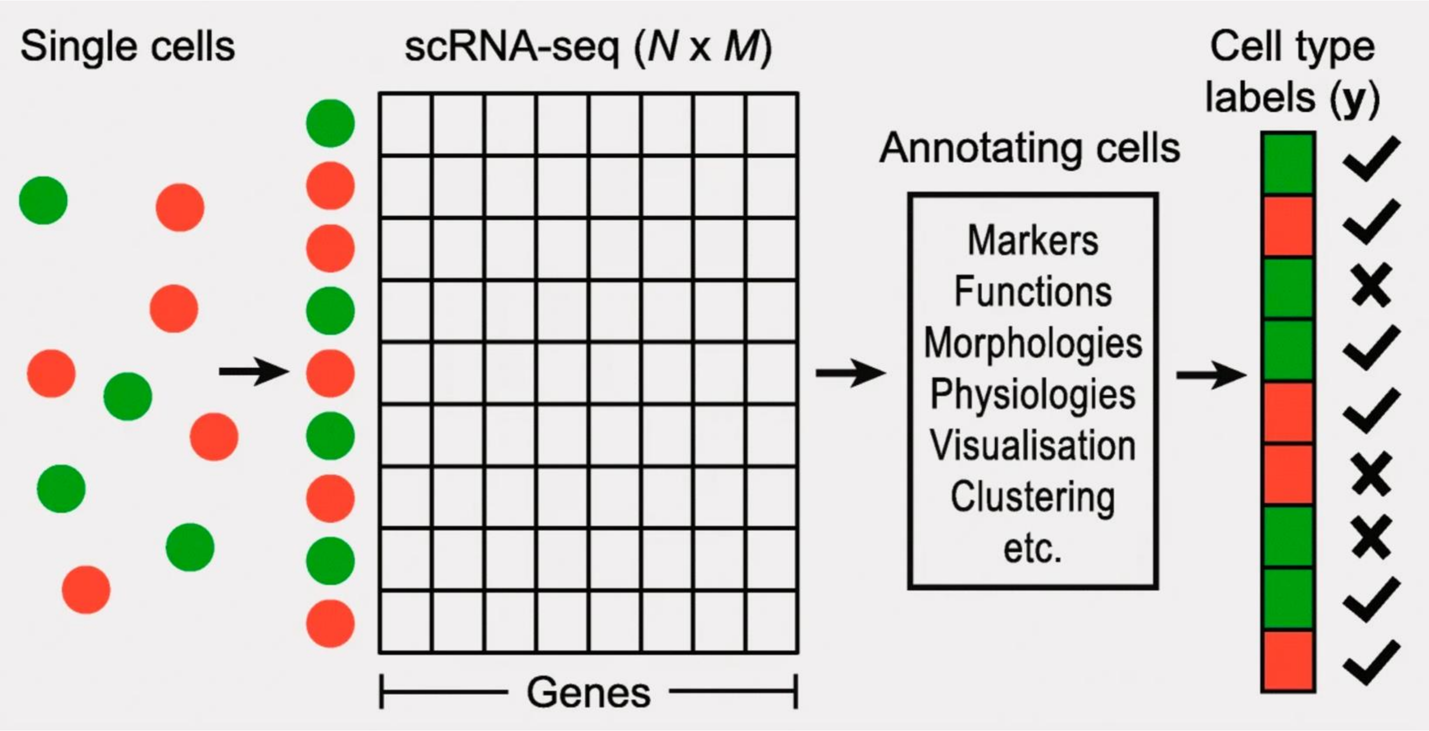


Image courtesy: https://www.rna-seqblog.com/removal-of-batch-effects-from-single-cell-rna-seq/

Challenges of working with single-cell data

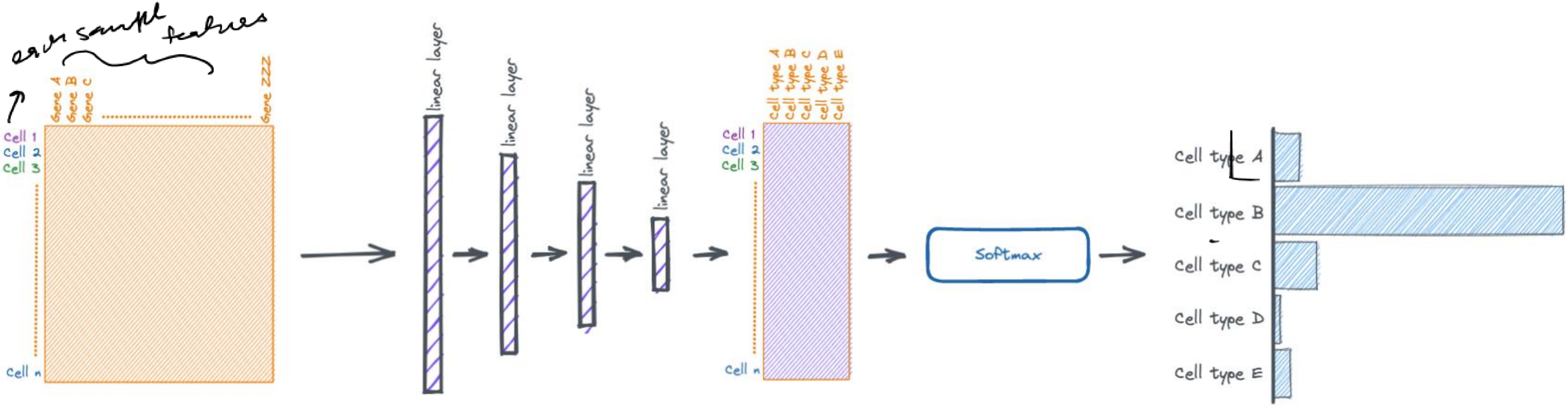
Can you train a deep learning model for this task?

- Downstream analysis:
  - Mapping to reference atlas



https://bmcgenomics.biomedcentral.com/articles/10.1186/s12864-019-6305-x/figures/1

Cell-type classification using deep learning



https://storrs.io/sc-deep-learning-explainer/

Challenges of working with single-cell data

Any questions?

- Data
  - Sparsity, errors and missing data
  - "Nuisance factors": Batch effects, etc.
- Downstream analysis:
  - Mapping to reference atlas
  - Integrating measurements

Review | Open Access | Published: 07 February 2020

Eleven grand challenges in single-cell data science

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Recap – Learn about deep CNN architectures and new applications

(1) Deep CNN architectures

(2) Deep CNNs in health applications

(3) Class activity: CNNs for genomics application

(4) New dataset: Single-cell gene expression

Wrap up

What was the clearest point today?

What was the muddiest point today?